AI-POWERED SMART GYM SYSTEM

Requirements Engineering

Functional Requirements · Non-Functional Requirements · User Requirements

A localized, bilingual smart-fitness platform combining real-time on-device posture AI, an affordable ESP32 IoT wearable, and Egyptian-first nutrition guidance.

Mobile application and computer vision: **Flutter**
Database: **Google Cloud Firestore**
Wearable: **ESP32 + MAX30102 + MPU6500 over Bluetooth Low Energy**

1. Introduction

1.1 Purpose

This document specifies the requirements for the AI-Powered Smart Gym System. It is intended to be the single agreed statement of what the system must do, how well it must do it, and what boundaries the design must respect. It serves the development team as the basis for design and implementation, the testing effort as the basis for verification, and the academic assessment as the record of engineering intent.

The document is written to be verifiable. Every functional requirement carries an acceptance criterion that can be demonstrated. Every non-functional requirement carries a quantified target, a verification method, and a reference identifier pointing to the published measurement, standard, specification or statute from which that number is drawn. Where the project deliberately sets a target stricter or looser than the cited figure, the entry says so and explains why.

1.2 Scope

The system is a bilingual Arabic and English mobile fitness platform for the Egyptian market. It unifies three capabilities that no reviewed competitor combines: real-time posture correction from the smartphone camera; biometric sensing from a low-cost custom wristband; and nutrition guidance built on Egyptian foods and Egyptian prices.

In scope:

- A cross-platform mobile application built in Flutter for Android and iOS.
- A computer-vision pose-coaching pipeline implemented inside that same Flutter application, running entirely on-device.

- An ESP32-based wristband with optical heart-rate sensing, inertial sensing (motion and step counting), battery-level monitoring and an on-board status LED, linked over Bluetooth Low Energy. The band derives heart rate, steps and an estimated calorie burn on-device, and receives the user's sex, age and weight from the phone.
- A Firestore-backed data tier holding accounts, workout history, biometric aggregates, meal logs and chat history.
- A thin backend orchestration service handling chatbot prompt assembly and provider failover.
- An Egyptian nutrition database with meal planning priced in Egyptian pounds.
- A domain-restricted bilingual conversational assistant.

Out of scope for this release:

- Any clinical, diagnostic or medical-monitoring function.
- Gym-side or operator-side software, membership management or class booking.
- Social networking, leaderboards or user-to-user messaging.
- Payment processing and subscription billing.
- Fall detection and emergency alerting, which are recorded as future work.

Throughout, "shall" denotes a binding requirement, "should" a strong recommendation, and "may" an option left to the implementer.

2.User Requirements

2.1 Stakeholders and User Classes

Class	Description	Primary concerns
Novice gym user	Trains without a coach; price-sensitive; prefers Arabic.	Safety, clear correction, low cost, natural Arabic.
Experienced lifter	Trains regularly; tracks load and volume.	Accurate rep counting, volume/1RM trends, low-friction logging.
Home trainee	No gym access; bodyweight/minimal equipment.	Full exercise coverage on a phone propped against a wall.
Nutrition-led user	Focused on eating well within an Egyptian budget.	Real Egyptian dishes, believable macros, EGP prices.
Development team	Six students in two sub-teams over one academic year.	Clear interfaces, testable requirements, a scope that fits the schedule.
Academic supervisors	Dr. Hania Hassan.	Engineering rigour, justified figures, honest treatment of limitations.

2.2 Novice Gym User

ID	The user shall be able to …	Realised by
UR-NOV-01	Receive real-time, plain-language correction on exercise form without needing a human coach.	FR-CV
UR-NOV-02	Train safely as a beginner, with warnings before risky exertion and no unexplained jargon.	FR-ALRT
UR-NOV-03	Interact with the app and coaching cues in Arabic that sounds natural, not machine-translated.	FR-LOC, FR-CHAT
UR-NOV-04	Get the full coaching experience without buying expensive equipment.	C-COST, C-TECH-09

2.3 Experienced Lifter

ID	The user shall be able to …	Realised by
UR-EXP-01	See accurate, automatic repetition counts and a form score during training.	FR-CV
UR-EXP-02	Log workouts quickly, including while offline in a gym with poor signal.	FR-DATA-03, NFR-REL-02

2.4 Home Trainee (optional when adding new exercises)

ID	The user shall be able to …	Realised by
UR-HOME-01	Train effectively with bodyweight or minimal-equipment exercises using only a propped phone.	FR-CV-01, FR-CV-12
UR-HOME-02	Receive a plan adapted to declared equipment access (home / bodyweight only).	FR-PROF-01

2.5 Nutrition-Led User

ID	The user shall be able to …	Realised by
UR-NUT-01	Search for and log real Egyptian dishes, by name, in Arabic or English.	FR-NUT-05
UR-NUT-02	Receive a daily meal plan built from Egyptian foods, priced in EGP, that meets their targets.	FR-NUT-05, -06
UR-NUT-03	Exclude foods they cannot or do not eat, and trust a plan will never include them.	FR-NUT-09

3. Functional Requirements

This section specifies what the system must do. Requirements are grouped by subsystem. Each carries an acceptance criterion stated in terms that can be demonstrated in a test or a review, so that “done” is never a matter of opinion.

3.1 Account, Authentication and Identity Management (FR-AUTH)

Identity is handled by Firebase Authentication; every authenticated principal is mirrored by exactly one Firestore profile document at /users/{uid}, where {uid} is the Firebase Auth UID. All rules in this group are enforced twice — once in the Flutter client for user experience, and once in Firestore Security Rules as the authoritative check.

ID	Requirements	Priority
FR-AUTH-01	The system shall allow a new user to create an account using an email address and a password through Firebase Authentication, and shall automatically create the corresponding Firestore profile document at /users/{uid} carrying the fields: uid, email, displayName, isMale, weight, age, and createdAt (server timestamp), height, experience, goal, training days, activityLevel.	Must
FR-AUTH-02	The system shall support sign-in via (a) email and password and (b) Google Sign-In as a federated provider, with both paths resolving to the same Firebase UID when the verified email address matches.	Must
FR-AUTH-03	The system shall provide a self-service password reset flow that sends a time-limited reset link to the registered email address without revealing whether that address exists in the system.	Must

ID	Requirements	Priority
FR-AUTH-04	The system shall enforce a password policy of at least 8 characters, shall accept passwords of at least 32 characters without truncation, shall permit all printable ASCII characters and the space character, and shall never store passwords or password hashes in Firestore — hashing and storage of credentials are handled entirely by Firebase Authentication.	Must
FR-AUTH-05	The system shall persist the authenticated session across application restarts and shall refresh the ID token silently in the background, prompting for re-authentication only when the refresh token is revoked or expired.	Must
FR-AUTH-06	On sign-out, the system shall clear the local isar database cache of all personally identifiable data, all cached biometric samples, and all cached chat history, while retaining only non-personal reference data such as the exercise catalogue.	Should
FR-AUTH-07	The system shall allow a user to permanently delete their account, which shall recursively delete the /users/{uid} document tree and the Firebase Auth record.	Should

3.2 User Profile, Onboarding and Goal Setting (FR-PROF)

Supplies the personalisation inputs consumed by coaching, nutrition and chat; derived values are recomputed client-side and written back to Firestore.

ID	Requirement	Pr.
FR-PROF-01	Collect 8 onboarding attributes (age, sex, height, weight, experience, goal, training days, activityLevel) in ≤ 8 screens.	Must
FR-PROF-02	Compute and store BMI, refreshed on every height/mass update.	Must
FR-PROF-03	Estimate Resting Energy Expenditure via the Mifflin-St Jeor equation.	Must
FR-PROF-04	Derive TDEE and protein/carb/fat targets from REE, activity factor and goal.	Must
FR-PROF-05	Estimate HRmax (208 − 0.7×age) and derive five training-intensity zones.	Should
FR-PROF-06	Set a default weekly activity target depending on activityLevel and goal (ex: 150 min, optional 300 min stretch, ≥ 2 strength days), user-overridable.	Should
FR-PROF-07	Let the user set/change interface language (Arabic/English) without restart.	Must

3.3 Computer Vision and On-Device Pose Coaching (FR-CV)

The entire CV pipeline runs inside the Flutter app; no frame, image or landmark stream ever leaves the handset.

ID	Requirement	Pr.
FR-CV-01	Acquire a ≥ 720p/30fps live camera stream from camera.	Must
FR-CV-02	Run BlazePose entirely on-device, extracting 33 landmarks (x, y, z, visibility, presence) + world landmarks per frame.	Must
FR-CV-03	Support Lite/Full/Heavy model variants, auto-selected by device capability tier, with manual override.	Must
FR-CV-04	Compute joint angles in Dart (knee, hip, trunk, elbow, shoulder) via 2-D vector trigonometry.	Must
FR-CV-05	Count repetitions with a per-exercise using hysteresis thresholds.	Must
FR-CV-06	Score posture risk against a per-exercise rule set (knee valgus, depth/ROM, lumbar flexion, forward lean, elbow flare, asymmetry).	Must
FR-CV-07	Deliver real-time corrective feedback via synchronised TTS+on-screen overlay, rate-limited to 1 cue / 2s.	Must
FR-CV-08	Render a colour-coded (green/amber/red) skeleton overlay on the camera preview.	Must
FR-CV-09	Discard low-visibility landmarks; suspend scoring and prompt repositioning when required landmarks are missing > 1s.	Must
FR-CV-10	Perform a pre-set framing calibration with a 3-second audible countdown before scoring begins.	Must
FR-CV-11	Never transmit or persist raw frames, images, video or per-frame landmarks — only derived session summaries.	Must
FR-CV-12	Ship a v1 exercise library of ≥ 3 movements (squat, deadlift, Bench press).	Must
FR-CV-13	Monitor inference frame rate; auto-downgrade model/resolution if it falls below the NFR-PERF-01 floor for > 3s.	Should
FR-CV-14	Produce a per-set summary: reps, time-under-tension, ROM, tempo, form score (0–100), fault counts.	Must

3.4 Data Model, Persistence and Synchronisation (FR-DATA)

Cloud Firestore is the system of record; the model is deliberately denormalised so every screen is servable from a small number of reads.

ID	Requirement	Pr.
FR-DATA-01	Implement the defined Firestore collection structure (/users, /workouts, /sets, /biometrics, /meals, /chat, /foods, /exercises, /appConfig, /auditLog).	Must
FR-DATA-02	Every document carries a schemaVersion integer and a serverTimestamp updatedAt.	Must
FR-DATA-03	Enable Firestore offline persistence — reads from cache, writes queued durably offline.	Must
FR-DATA-04	Aggregate biometric samples into fixed time-window documents, not one document per sample.	Must
FR-DATA-05	Use batched writes/transactions for multi-document updates, within the platform's per-commit limit.	Must
FR-DATA-06	Security Rules deny by default; a client accesses only its own /users/{uid} subtree; catalogues are read-all / admin-write.	Must
FR-DATA-07	Mirror food/exercise catalogues into local storage at launch, refreshed on version mismatch, for offline search.	Should
FR-DATA-08	Define composite indexes in advance for every query, within the project's index quota.	Should

3.5 Egyptian Nutrition Database and Meal Planning (FR-NUT)

The project's principal localisation asset: food entries carry bilingual names, macros and EGP prices.

ID	Requirement	Pr.
FR-NUT-01	Ship a curated Egyptian food catalogue of ≥ 300 entries covering named staples (ful, koshary, molokhia, etc.), sourced from a dataset to be finalised consistent with NFR-DQ-05's sourcing preference.	Must
FR-NUT-02	Each food document carries bilingual names, category, and macros per 100 g plus ≥ 1 household portion.	Must
FR-NUT-03	Each food carries an indicative EGP price with a priceAsOf date; prices > 90 days are marked stale.	Should
FR-NUT-04	Food search accepts Arabic, English and transliterated spellings with fuzzy matching.	Must
FR-NUT-05	Generate a daily meal plan (3 meals + ≤ 2 snacks) meeting TDEE within 5% and protein within 10%.	Must
FR-NUT-06	Show estimated plan cost in EGP; allow regeneration under a stated daily budget ceiling.	Should
FR-NUT-07	Allow logging consumed foods (search / portion / repeat-yesterday) with running daily totals.	Must
FR-NUT-08	Support a fasting mode with a user-defined eating window, incl. a Ramadan (iftar/suhoor) preset.	Could
FR-NUT-09	Record dietary exclusions and never place an excluded item in a generated plan.	Must

3.6 Bilingual Conversational Assistant (FR-CHAT)

ID	Requirement	Pr.
FR-CHAT-01	Understand/reply in Modern Standard Arabic, Egyptian, Arabic and English, matching the user's message language.	Must
FR-CHAT-02	Route completions to a primary LLM provider with automatic failover to a secondary on error/timeout/rate-limit.	Should
FR-CHAT-03	Constrain the assistant to fitness/nutrition/safety domains; decline out-of-domain requests politely.	Must
FR-CHAT-04	Inject goal, experience, limitations, recent workout summary and macro targets as context; never send raw biometric series.	Must
FR-CHAT-05	Stream the reply to the client.	Should
FR-CHAT-06	Refuse medical diagnosis / prescription / clinical interpretation of biometrics; show a persistent non-diagnostic disclaimer.	Must
FR-CHAT-07	Persist chat history (role, content, language, model, tokens, timestamp) under /users/{uid}/chat; allow clearing it.	Must
FR-CHAT-08	Return a cached/templated fallback for common intents if both providers are unavailable.	Should

3.7 Session Management, Alerts and Safety (FR-ALRT)

Covers how a workout is structured and what the system tells the user while it is happening.

ID	Requirement	Pr.
FR-ALRT-01	Model a workout as an ordered exercise list with sets/reps/load, tracking per-set completion.	Must
FR-ALRT-02	Provide an automatic inter-set rest timer.	Must
FR-ALRT-03	Raise a caution when heart rate exceeds 90% of estimated HRmax; recommend reducing intensity.	Should
FR-ALRT-04	Show a warm-up recommendation before the first working set and a mandatory first-session safety acknowledgement.	Should

3.8 Localisation, Interface and Accessibility (FR-LOC)

Arabic support is a first-class functional requirement, not a translation layer — the project's stated differentiator.

ID	Requirement	Pr.
FR-LOC-01	Provide a complete Arabic + English UI.	Must
FR-LOC-02	Fully mirror layout (nav, icons, charts, lists) in Arabic via RTL directionality.	Must
FR-LOC-03	Render numerals in the chosen numeral system; format dates/time/currency to the Egyptian locale (EGP, metric).	Must
FR-LOC-04	Provide all spoken coaching cues in both languages, Arabic in Egyptian phrasing.	Should
FR-LOC-05	Show localised, actionable error messages; never surface raw exceptions or stack traces.	Must

3.9 Progress Analytics and Reporting (FR-ANL)

Computed from set summaries and biometric aggregates already stored; no additional raw data is collected.

ID	Requirement	Pr.
FR-ANL-03	Chart mean form score per exercise over time.	Should
FR-ANL-04	Chart HR trends per session and resting HR over time, from stored aggregates.	Should
FR-ANL-05	Track adherence (sessions completed vs. planned; minutes vs. weekly target).	Should

4. Non-Functional Requirements

This section specifies how well the system must perform. Each row states the quantified target; “Target” in the reference column marks a figure set by the team's own engineering judgement rather than a published measurement. Full justifications and verification methods are maintained in the project's verification records; reference identifiers resolve in Section 7.

4.1 Performance and Responsiveness (NFR-PERF)

ID	Quantified target	Ref
NFR-PERF-01	On-device pose inference ≤ 30 ms/frame, sustaining ≥ 25 fps during an active set.	R1,R2
NFR-PERF-02	Feedback latency: ≤ 100 ms to on-screen overlay; ≤ 300 ms to the spoken cue.	R4
NFR-PERF-04	The band notifies heart-rate/calorie, motion/step and battery data every 2 s; each update is visible in the UI within 200 ms of being received by the phone.	R10,R11
NFR-PERF-05	Writes acknowledged locally at once; committed to Firestore within 500 ms (p95) on 4G.	R4,R8
NFR-PERF-06	A cold query for the 30 most recent workouts completes within 1 s (p95).	R4,R8
NFR-PERF-07	Chat reply begins streaming within 1.5 s (p95); a 150-token reply completes within 3 s.	R4,R18
NFR-PERF-08	App start ≤ 3 s cold, ≤ 1.5 s warm, ≤ 1 s hot.	R13
NFR-PERF-09	Local food search returns ranked results within 500 ms of the last keystroke (≤ 1,000 entries).	R4
NFR-PERF-10	Camera preview latency (movement to screen) ≤ 100 ms.	R4
NFR-PERF-11	Optical sensor sampled at ≈ 100 sps (effective) while the band is connected.	R12
NFR-PERF-12	Biometric data committed as one aggregated document per 60 s window per user, never per sample.	R6,R9
NFR-PERF-13	Application binary ≤ 60 MB (Android bundle) including the bundled pose model.	Target

4.2 Reliability, Availability and Fault Tolerance (NFR-REL)

ID	Quantified target	Ref
NFR-REL-01	Cloud data tier ≥ 99.9% monthly availability.	R7
NFR-REL-02	Fully usable offline (coaching, set logging, rest timer, food logging), zero data loss, auto-reconcile on reconnect.	R6
NFR-REL-04	Wristband link re-establishes within 10 s of returning to range; supervision timeout per BLE spec.	R10
NFR-REL-05	Failover to the secondary LLM provider if no first token within 5 s from the primary.	R18
NFR-REL-09	No single failure (band, network, either LLM provider, camera permission) cascades — each degrades only its own feature.	Target

4.3 Capacity, Scalability and Cost (NFR-SCAL)

Targets are set against free/low-cost tiers so the project is demonstrable on a student budget.

ID	Quantified target	Ref
NFR-SCAL-01	Support 5,000 registered accounts / 500 daily active users without architectural change.	Target
NFR-SCAL-02	At 500 DAU, operate inside the Firestore free tier (50k reads / 20k writes / 20k deletes per day); ≈ 100 reads, 40 writes per user per day.	R6
NFR-SCAL-04	No unbounded document growth; unbounded lists are modelled as subcollections, within the 1 MiB document limit.	R6
NFR-SCAL-06	LLM spend ceiling: USD 0.25 / active user / month (≈ USD 100/month at 500 DAU).	R18
NFR-SCAL-07	Food catalogue scalable to 2,000 entries, exercise catalogue to 100, without search-latency degradation.	Target
NFR-SCAL-08	Chat history capped per user; oldest messages archived/pruned to bound storage growth.	R6

4.4 Security and Privacy (NFR-SEC)

The system processes health-related data, which Egyptian law treats as a special category attracting enhanced protection.

ID	Quantified target	Ref
NFR-SEC-01	TLS ≥ 1.2 in transit; platform default encryption at rest.	R6
NFR-SEC-02	Security Rules deny-by-default, covered by an automated test suite incl. cross-user and privilege-escalation cases.	R6
NFR-SEC-03	Password handling: ≥ 8 char min, accept ≥ 32 chars, no truncation, all printable ASCII + space, no hints stored, compromised-password check.	R15
NFR-SEC-04	No raw camera frame/image/video ever leaves the device, incl. crash/diagnostic uploads; biometrics leave only as aggregates, post-consent.	R16
NFR-SEC-05	No third-party API credentials embedded client-side; all LLM calls proxied through the backend.	Target
NFR-SEC-06	Chat text stripped of direct identifiers before transmission; user told in the consent flow that content goes to an external provider.	R16
NFR-SEC-07	Biometric window documents retained ≤ 12 months, then aggregates only; stated in the privacy notice.	R16
NFR-SEC-08	Database region documented; any cross-border transfer disclosed in the privacy notice.	R16
NFR-SEC-09	A documented breach-response procedure (detection, assessment, notification) must exist.	R16
NFR-SEC-11	Chat completions rate-limited server-side, per user and per IP.	R18
NFR-SEC-12	All admin actions on shared catalogues written to an append-only, user-unwritable audit log.	R16

4.5 Usability and Accessibility (NFR-USE)

ID	Quantified target	Ref
NFR-USE-03	Starting a tracked set from the home screen takes ≤ 3 taps.	Target
NFR-USE-04	Text contrast ≥ 4.5:1, both themes/languages.	R23
NFR-USE-05	Every pointer target ≥ 44×44 logical pixels.	R23
NFR-USE-06	Arabic UI judged natural (not literal translation) by ≥ 4 native reviewers.	Target
NFR-USE-07	Coaching cues intelligible over typical gym noise; every audio cue has a visual counterpart.	Target
NFR-USE-08	No screen requires reading more than 40 words to complete its primary task.	Target

4.6 Compatibility and Portability (NFR-COMP)

ID	Quantified target	Ref
NFR-COMP-01	One Flutter codebase; functional parity between Android and iOS, documented exceptions only.	Target
NFR-COMP-02	Support Android 8.0+ (API 26) and iOS 13+.	Target
NFR-COMP-03	Pose coaching requires ≥ 3 GB RAM + 720p/30fps camera; cleanly disabled below spec.	R2
NFR-COMP-04	Wristband link requires BLE 4.2+; its absence disables only biometric features.	R10

4.7 Maintainability, Testability and Supportability (NFR-MNT)

ID	Quantified target	Ref
NFR-MNT-01	Layered architecture (presentation/domain/data); domain layer has no dependency on Firestore, camera, or any plugin.	Target
NFR-MNT-02	Domain-layer unit-test coverage ≥ 70%.	Target
NFR-MNT-03	Exercise rules/cues are declarative configuration, not branching code — new exercises need no engine change.	Target
NFR-MNT-05	Reproducible build from a clean checkout, one documented command per component (app, backend, firmware).	Target
NFR-MNT-06	Diagnostic logging is structured, disabled by default in release builds, and never logs personal/biometric data or credentials.	R16

4.8 Power, Thermal and Hardware Efficiency (NFR-PWR)

ID	Quantified target	Ref
NFR-PWR-01	Wristband: ≥ 8 h continuous session use on one charge.	Target
NFR-PWR-05	No sustained thermal throttling in a 45-min session; app downgrades model/rate before the platform throttles.	R2
NFR-PWR-06	Wristband enters deep sleep when the battery voltage stays at or below 2.95 V for 5 consecutive readings, to protect the cell.	Target

4.9 Data Quality and Accuracy (NFR-DQ)

ID	Quantified target	Ref
NFR-DQ-01	Reported HR agrees with a chest-strap reference within ±10 bpm for ≥ 90% of paired samples.	R12
NFR-DQ-03	Repetition counting ≥ 95% accurate, fault classification ≥ 90% accurate, vs. labelled ground truth (3 v1 exercises).	R21
NFR-DQ-04	≥ 95% of catalogue entries have a complete macro set; 100% carry a source reference.	R19
NFR-DQ-05	Indigenous dishes sourced from a national composition table in preference to a generic international one.	R19

4.10 Legal, Ethical and Compliance Attributes (NFR-LEG)

ID	Quantified target	Ref
NFR-LEG-01	App states prominently (onboarding + settings) that it is a wellness aid, not a medical device.	R16
NFR-LEG-02	Data processing rests on explicit, per-purpose consent; user can access/correct/erase/object/withdraw.	R16
NFR-LEG-03	Health data handled as a special category with enhanced safeguards.	R16
NFR-LEG-04	Third-party licence obligations honoured (incl. Apache-2.0 attribution); an open-source notices screen ships.	R22
NFR-LEG-05	Coaching/nutrition guidance stays within general-wellness advice; declines to advise on injuries/conditions beyond recommending professional assessment.	R16,R17
NFR-LEG-06	Activity recommendations stay within WHO guidance (150–300 min moderate / 75–150 min vigorous per week + strength days).	R17

5. Constraints

A constraint is a boundary the design cannot cross — imposed by a mandated technology decision, a platform limit, a law, a budget, a schedule or the physical environment, rather than something the team chooses to achieve.

5.1 Technology and Architecture (C-TECH)

ID	Constraint
C-TECH-01	App + entire CV pipeline built in Flutter/Dart, one codebase; the pose model is invoked via a platform-channel/FFI binding. (pose_detection) (https://pub.dev/packages/pose_detection)
C-TECH-02	Cloud Firestore is the sole system of record — no relational database, no server-side joins.
C-TECH-03	Pose inference is on-device only — no cloud vision service, no frame upload.
C-TECH-04	Wristband↔handset link is BLE only — no Wi-Fi data path on the ESP32.
C-TECH-05	Conversational AI is a hosted third-party LLM API — no self-hosted or fine-tuned model.
C-TECH-06	A local embedded database caches reference data on-device for offline operation.
C-TECH-07	The smartphone camera is the only vision sensor.

5.2 Platform and Runtime (C-PLAT)

ID	Constraint
C-PLAT-01	BLE connection interval is 7.5 ms–4 s in 1.25 ms steps (Android ≥ 7.5 ms, iOS ≥ 15 ms).
C-PLAT-02	Platform vitals tooling flags startup as excessive at 5 s / 2 s / 1.5 s (cold/warm/hot).
C-PLAT-03	Camera and body-sensor permissions are runtime permissions the user may revoke at any time.

5.3 Firestore Platform Limits (C-DATA)

Hard platform limits (1 MiB max document size; 1,500-byte max indexed field value; 200 free composite indexes; 40,000 index entries/document; free tier = 1 GiB storage, 50k reads / 20k writes / 20k deletes per day, 10 GiB egress/month) drive four design rules:

ID	Constraint
C-DATA-01	A user's workout history is a subcollection of documents, never an array field — the 1 MiB document limit would eventually be breached.
C-DATA-02	Biometric samples are aggregated into per-window documents, not written per sample — the free tier allows only 20,000 writes/day.
C-DATA-03	Chat history is capped and pruned per user, the only unbounded-growth collection in the model.
C-DATA-04	Every queried/ordered field is kept under the 1,500-byte indexed-value limit, since longer values are silently truncated.

5.4 Operational and Environmental (C-ENV)

ID	Constraint
C-ENV-01	Gym lighting is uneven and often dim — comparable pipelines fall from ≈ 96% to ≈ 92% accuracy under it.
C-ENV-02	The handset is hand-held, not tripod-mounted — camera angle varies (≈ 94% accuracy at a 30° offset).
C-ENV-03	Gym equipment and other users partially obscure the body for many repetitions.
C-ENV-04	The wristband is worn under sweat, impact and high motion — degrades optical coupling, stresses the enclosure.
C-ENV-05	Mobile connectivity inside gyms is frequently poor or absent, particularly in basements.
C-ENV-06	Users interact mid-set, sweaty-handed, at arm's length, often in a noisy room.

6. References

Every numeric figure in Sections 5 and 6 traces to an entry below. Figures marked "Target" in Section 5 are the team's own engineering choices and are not listed here. Each entry links directly to its source.

ID	Source, the figures it supports, and a direct link
R1	Bazarevsky et al., "BlazePose: On-device Real-time Body Pose Tracking," CVPR Workshop, 2020 (arXiv:2006.10204). 33-landmark topology; model size; up to 31 fps (Pixel 2, Lite). https://arxiv.org/abs/2006.10204
R2	Google, "MediaPipe Pose — solution documentation." Per-frame latency (Pixel 3, GPU delegate): Lite 20 ms, Full 25 ms, Heavy 53 ms. https://mediapipe.readthedocs.io/en/latest/solutions/pose.html
R3	Google Research Blog, "On-device, Real-time Body Pose Tracking with MediaPipe BlazePose," 2020. Real-time CPU / super-real-time GPU performance; 33-point topology rationale. https://research.google/blog/on-device-real-time-body-pose-tracking-with-mediapipe-blazepose/
R4	J. Nielsen, "Response Times: The 3 Important Limits," Nielsen Norman Group. 0.1 s instantaneous, 1.0 s flow, 10 s attention thresholds. https://www.nngroup.com/articles/response-times-3-important-limits/
R6	Google Cloud, "Firestore — Quotas and limits." Free-tier and platform hard limits used throughout Sections 5 and 6. https://cloud.google.com/firestore/quotas
R7	Google Cloud, "Firestore SLA." 99.999% (multi-region) / 99.99% (regional) availability commitment. https://cloud.google.com/firestore/sla
R8	Google Cloud, "Firestore overview." Single-digit-millisecond read latency; multi-region replication. https://cloud.google.com/firestore/native/docs
R9	Google Cloud, "Firestore — Best practices." The 500/50/5 ramp-up rule. https://cloud.google.com/firestore/docs/best-practices

ID	Source, the figures it supports, and a direct link
R10	Silicon Labs / Microchip BLE connection-parameter guidance. Interval range, supervision-timeout formula, interval–current relationship. https://docs.silabs.com/bluetooth/2.13/bluetooth-general-system-and-performance/optimizing-current-consumption-in-bluetooth-low-energy-devices
R11	Punch Through, "BLE Connection Interval and Peripheral Latency." Android 7.5 ms vs. iOS 15 ms minimum interval. https://punchthrough.com/ble-connection-interval-throughput/
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R17	WHO Guidelines on Physical Activity and Sedentary Behaviour, 2020 (Bull et al., Br. J. Sports Med., 2020). Weekly activity minutes. https://www.who.int/publications/i/item/9789240015128
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ID	Source, the figures it supports, and a direct link
R21	“AI Trainer: Autoencoder-Based Approach for Squat Analysis and Correction,” IEEE Access, vol. 11, 2023, pp. 107135–107149. 94% squat-classification accuracy. https://doi.org/10.1109/ACCESS.2023.3316009
R22	Apache License 2.0. Licence terms for the pose-estimation runtime. https://www.apache.org/licenses/LICENSE-2.0
R23	W3C WCAG 2.1, SC 1.4.3 and SC 2.5.5. Contrast ratio and pointer target-size minimums. https://www.w3.org/WAI/WCAG21/
R24	Mifflin et al., “A new predictive equation for resting energy expenditure,” Am. J. Clin. Nutr., 51(2), 1990, pp. 241–247. REE equation. https://doi.org/10.1093/ajcn/51.2.241
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R28	Supporting literature carried forward from the project's reference list. Background support for Section 2. 10537312 · 10275938 · 9298212 · 10675520 · 10061134 · 9398455 · 11036040 · ChatDiet (DOI)